Lecture 12: DNN Interpretability---Input Interpretability (Cont.)

CS 580B1: Trustworthy Machine Learning

Fall 2024

Acknowledgments

Course slides derived from material created by Rajeev Alur and Osbert Bastani at UPenn (link to their course)

Other relevant courses on Trustworthy Machine Learning:

CS 329T at Stanford

CS 521 at UIUC

ECE1784H/CSC2559H at U Toronto

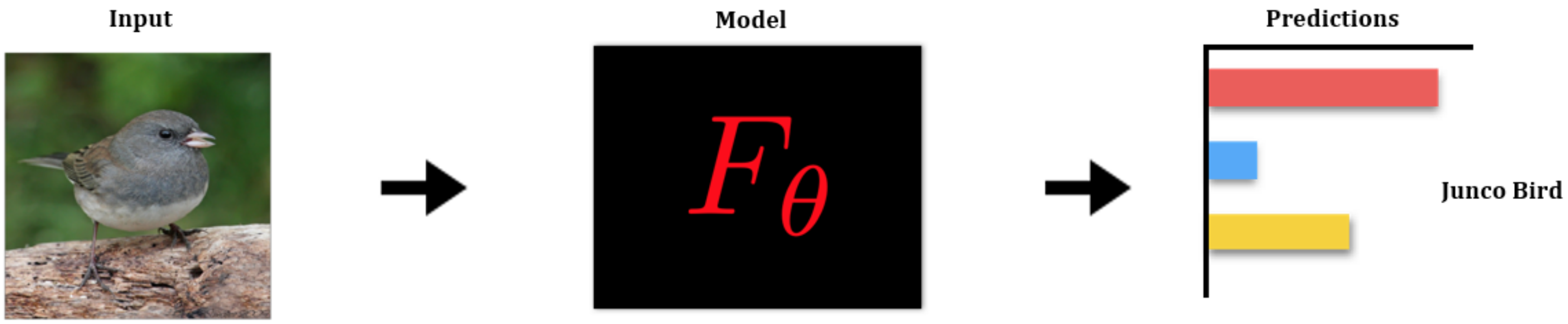
Types of post-hoc interpretability techniques (Recap)

- Input vs internal interpretability
- Local vs global
- Intrinsic vs post-hoc
- Model-specific vs model-agnostic (White-box vs black-box)
- …

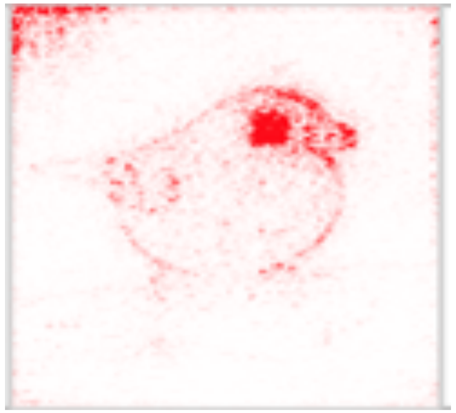
Feature attribution methods (Recap)

- Goal: Explain why the model made a particular prediction on a specific input
- Solution: Select/rank a subset of input features that contributed the most to model prediction
- Local explanations
- Model-agnostic vs Gradient-based
- Today:
  - **LIME:** Ribeiro et al., "Why should I trust you?" Explaining the predictions of any classifier, KDD'16
  - **Saliency Maps:** Simonyan et al., Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps, ICLR'14
  - **Integrated Gradients:** Sundararajan et al., Axiomatic Attribution for Deep Networks, ICML'17

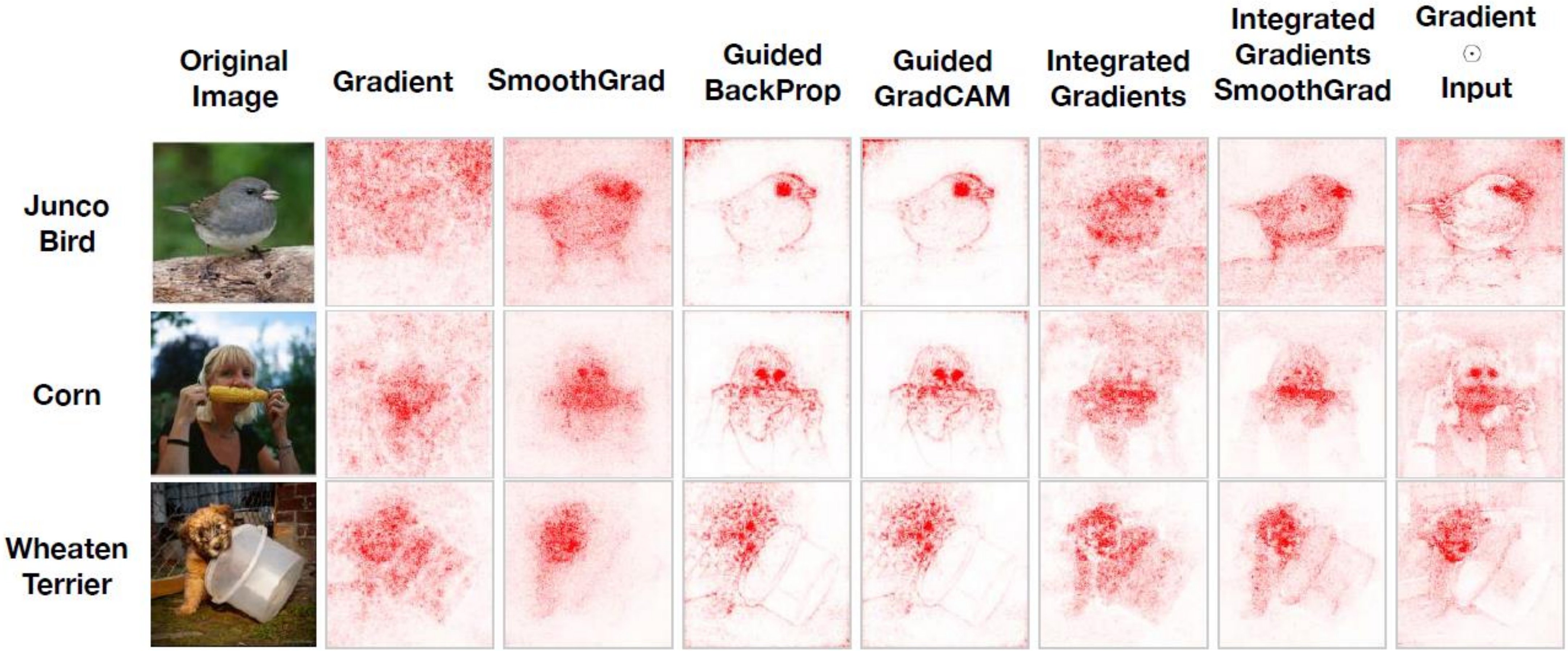
Saliency maps



What parts of the input are most relevant for the model's prediction: **'Junco Bird'?**

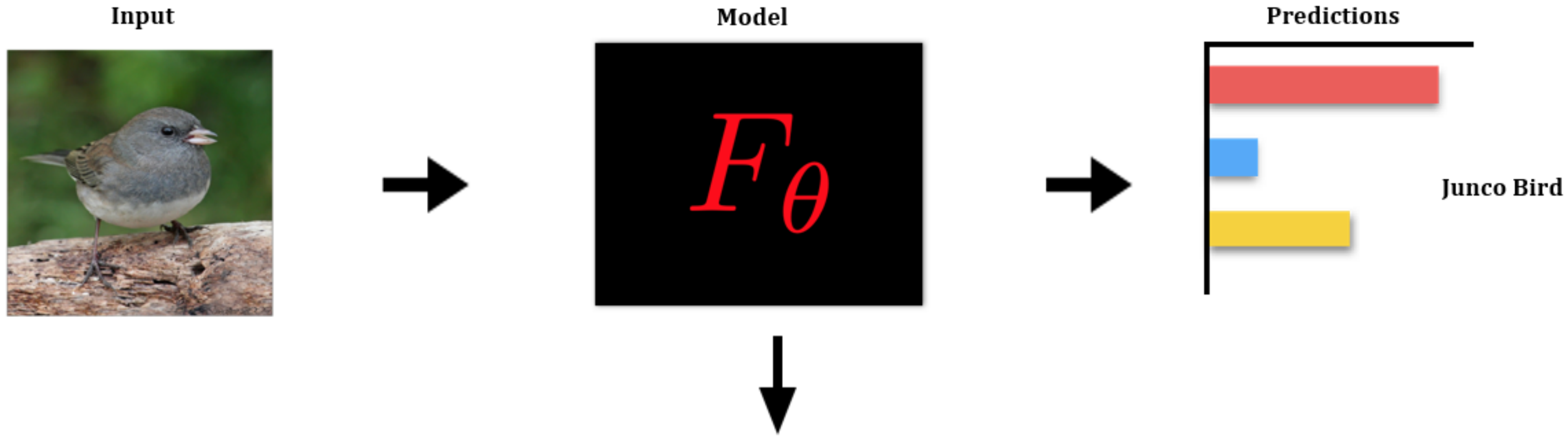


Different kinds of saliency maps



Adebayo et al., Sanity Checks for Saliency Maps, NeurIPS'18

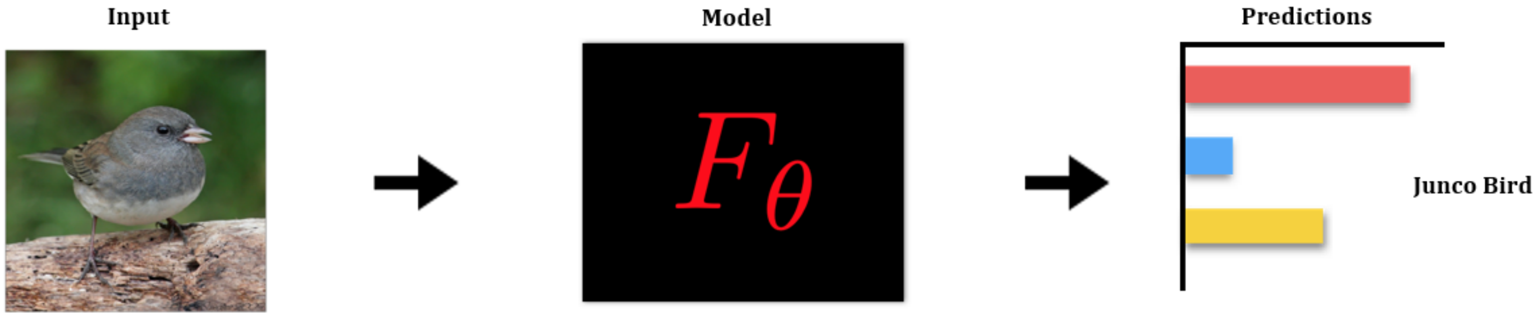
Saliency maps for DNNs



$$F : \mathbb{R}^d \rightarrow \mathbb{R}^c$$ **Model**

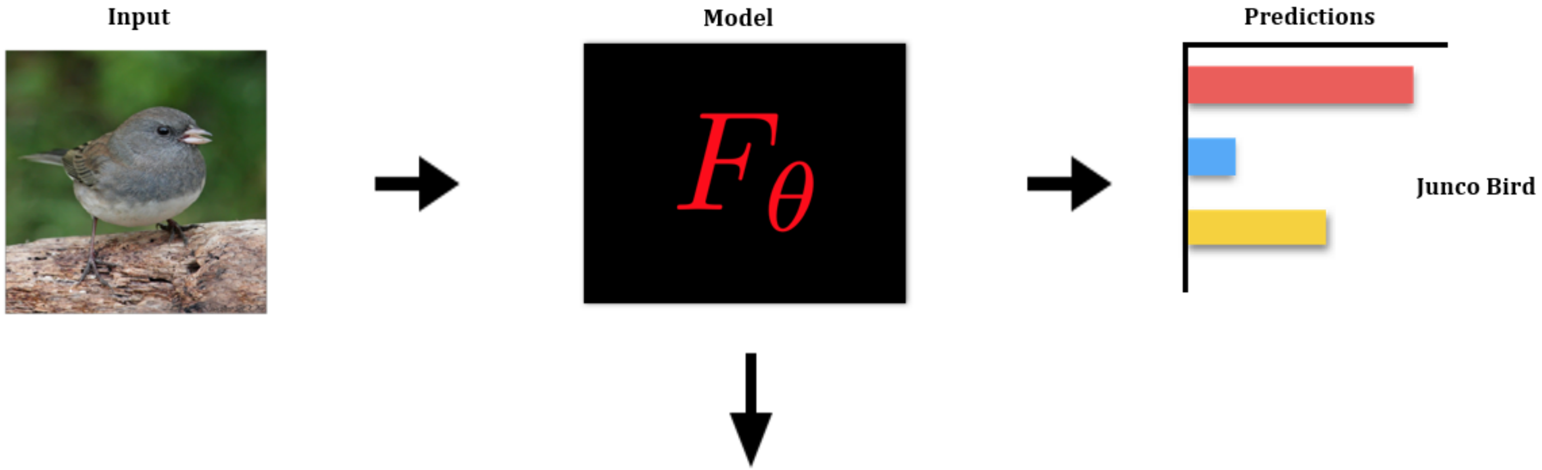
$$F_i : \mathbb{R}^d \rightarrow \mathbb{R}$$ **class specific logit**

Input gradient



- What’s the contribution of a pixel in input image to the prediction “Junco bird”?
- Solution: Compute the gradient of the output logit w.r.t. the pixel

Input gradient



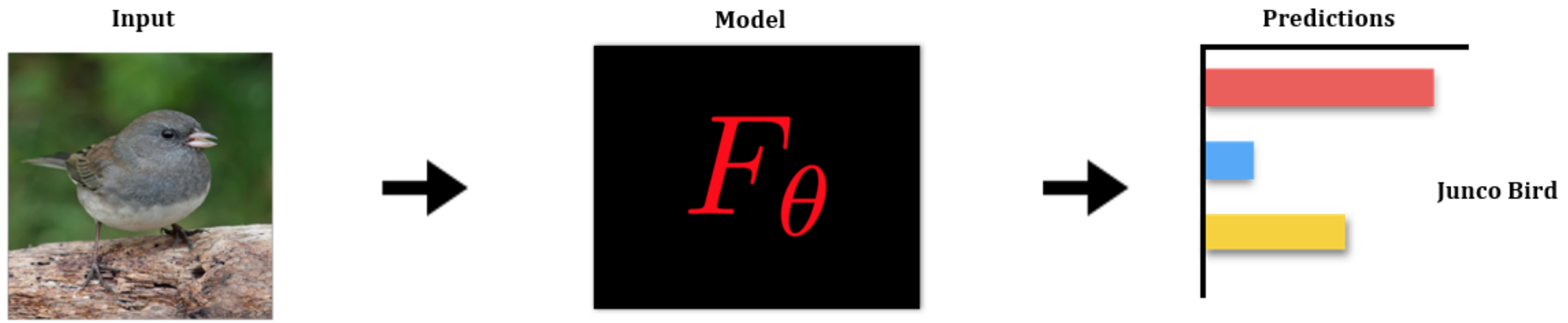
$$\nabla_x F_i(x) \quad \in \mathbb{R}^d$$

Logit

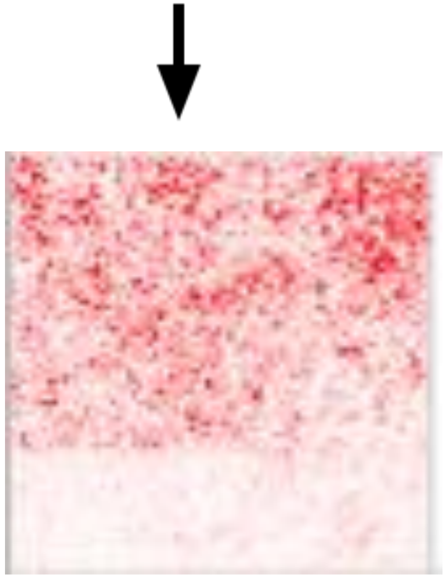
Input

Same dimension as the input

Input gradient



$$\nabla_x F_i(x)$$



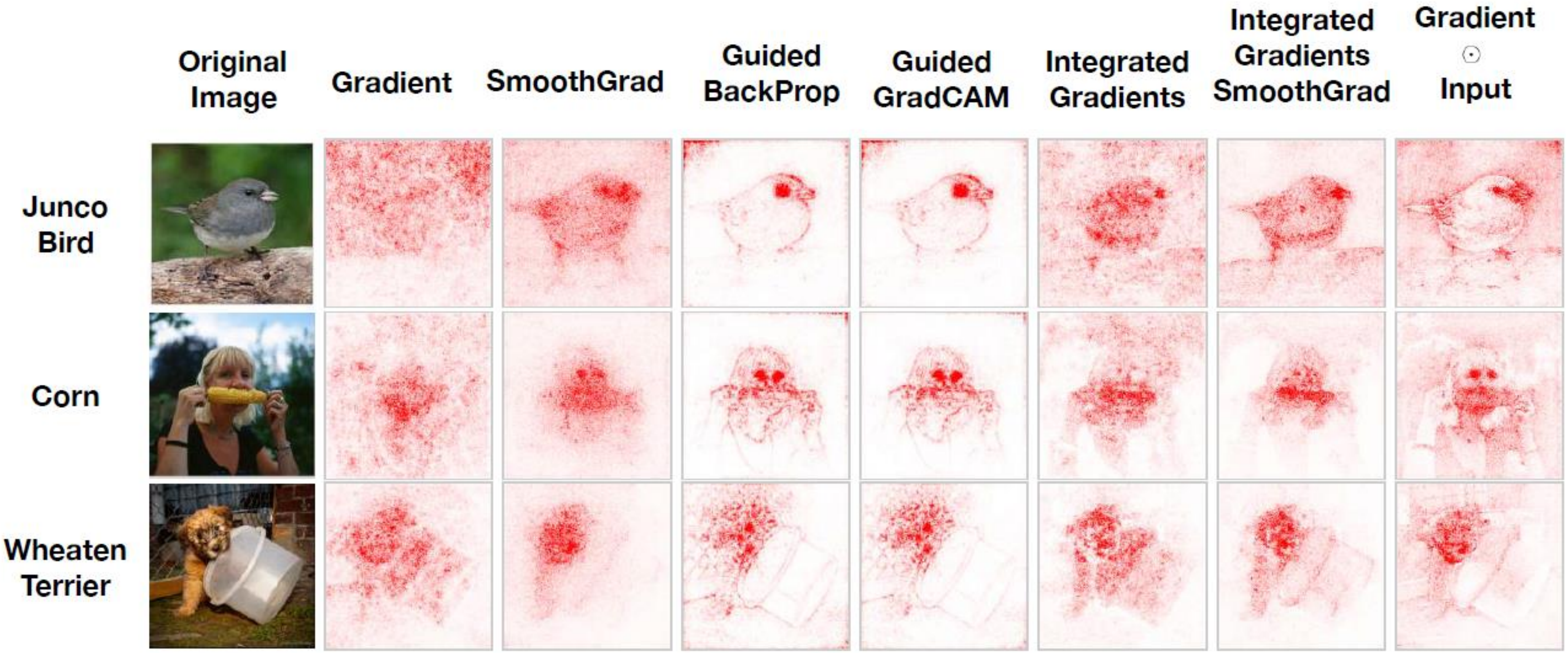
Visualize as heat map

Beyond input gradients

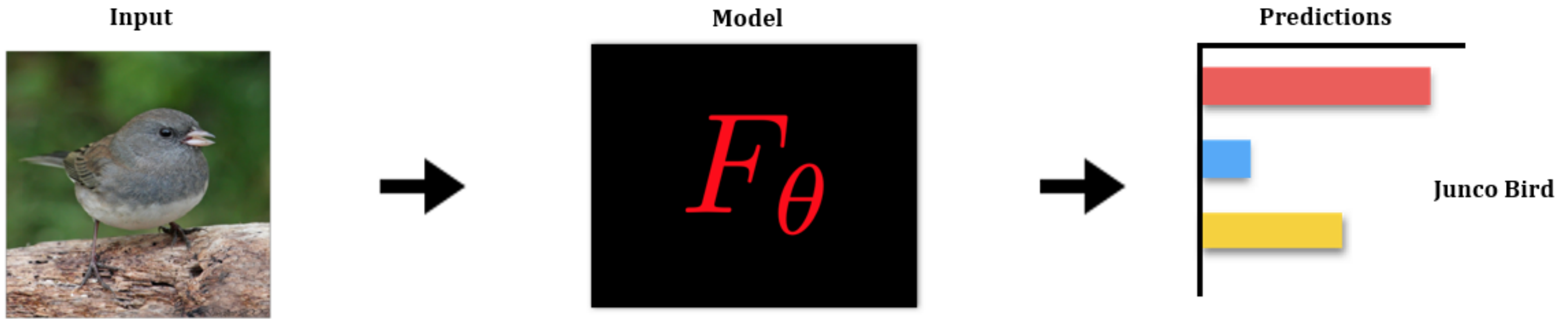
- Input gradients capture sensitivity to individual pixels/features well, but …raw gradients are visually noisy

- Many improvements proposed in literature

Beyond input gradients



SmoothGrad



SmoothGrad

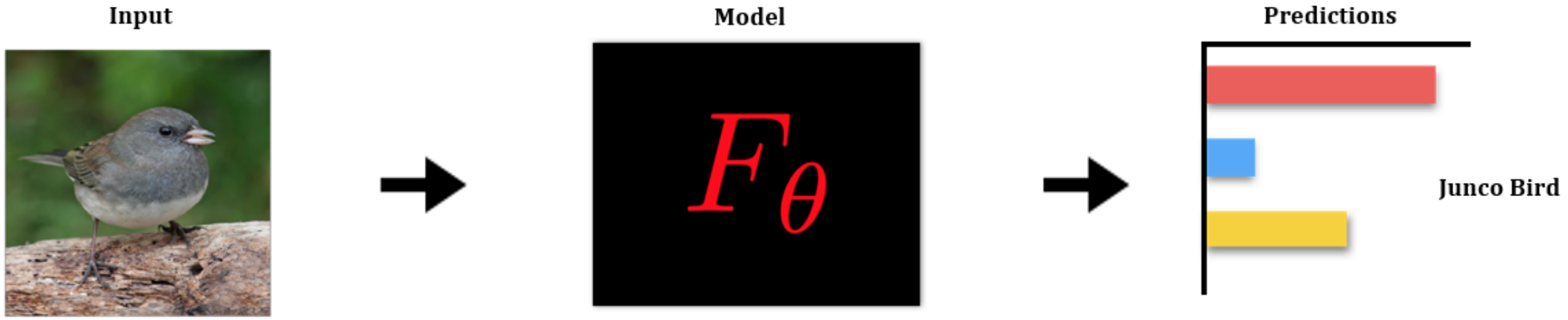
$$\frac{1}{N}\sum_{i}^{N}\nabla_{(x+\epsilon)}F_i(x+\epsilon)$$

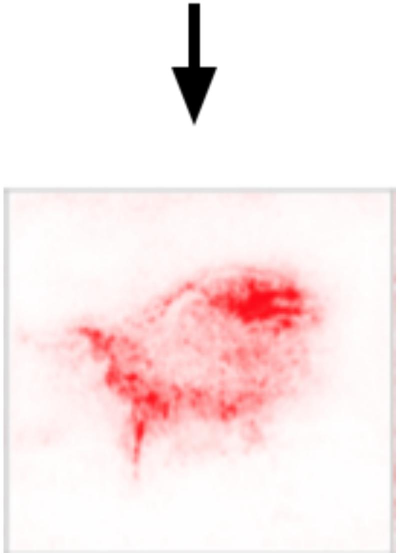
Gaussian noise

- Construct N perturbations of input x
- By adding noise ε sampled from Gaussian distribution with standard deviation σ
- For each variant compute input gradient
- Take average

Smilkov et al., SmoothGrad: removing noise by adding noise, 2017

SmoothGrad





$$\frac{1}{N}\sum_{i}^{N}\nabla_{(x+\epsilon)}F_i(x+\epsilon)$$

Average Input-gradients of “noisy” variants

SmoothGrad: Effect of noise

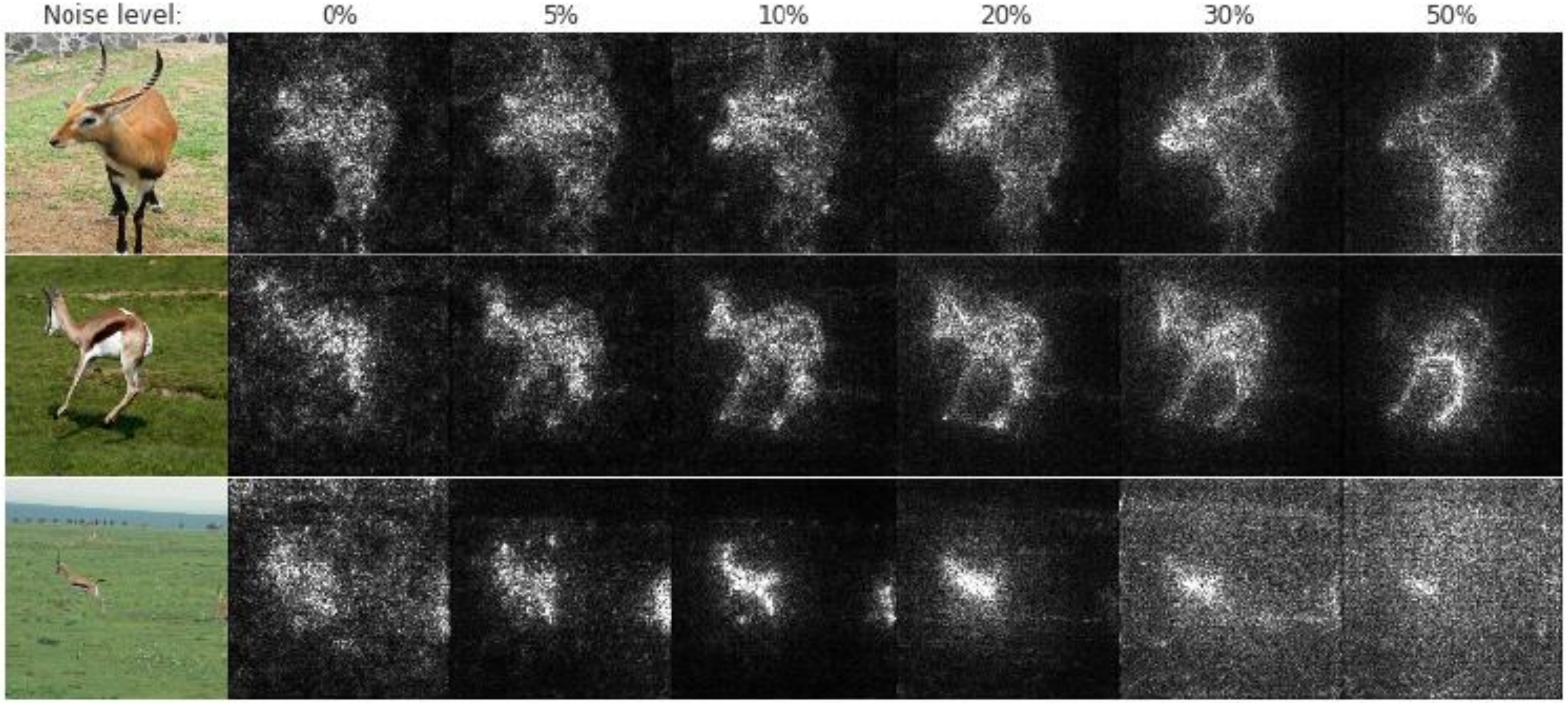
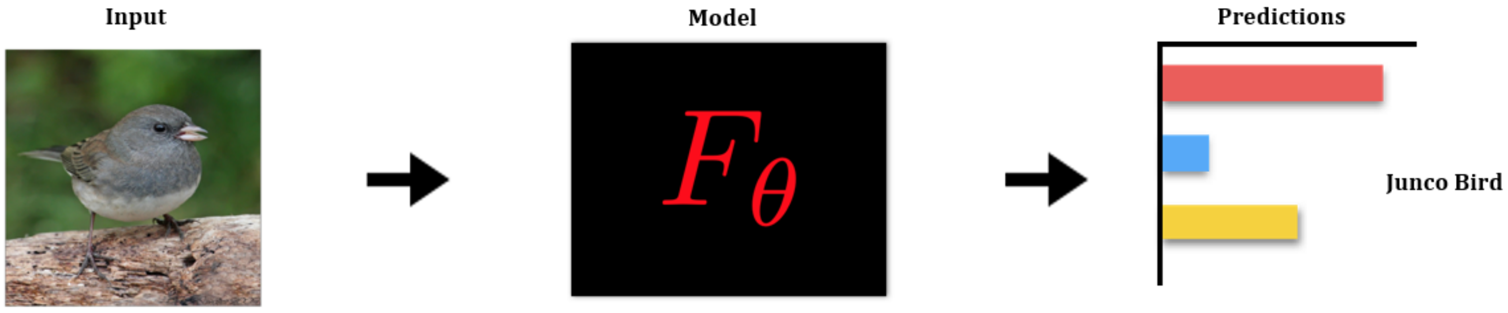


Figure 3. Effect of noise level (columns) on our method for 5 images of the gazelle class in ImageNet (rows). Each sensitivity map is obtained by applying Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the input pixels for 50 samples, and averaging them. The noise level corresponds to $\sigma/(x_{max} - x_{min})$.

Gradient-Input



$$\nabla_x F(x) \odot x$$

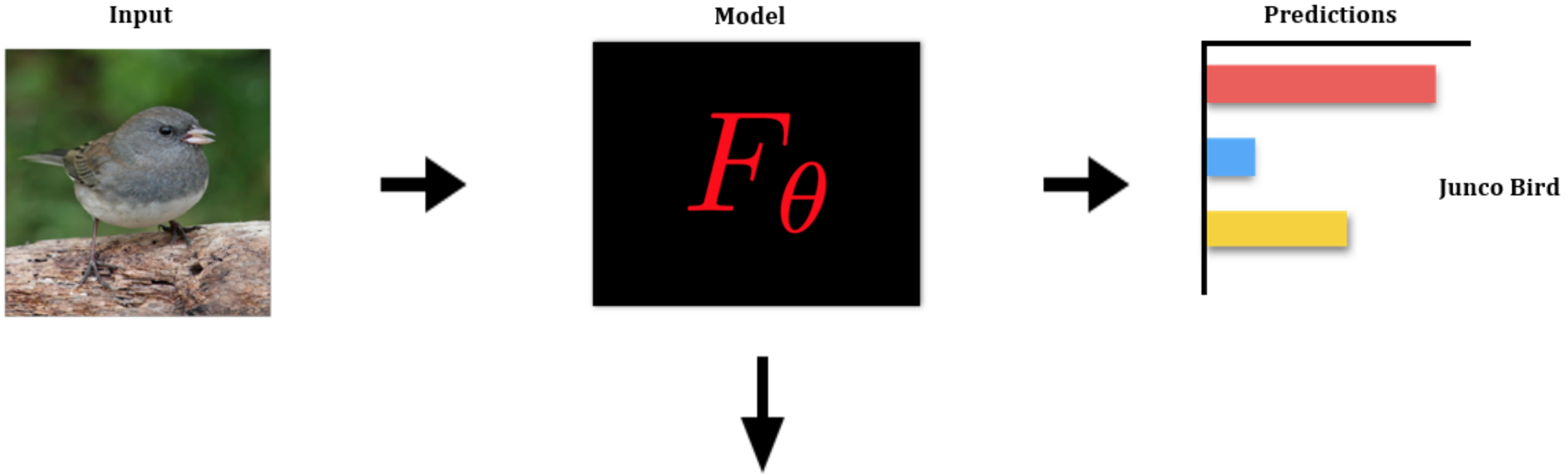
Input gradient

Input

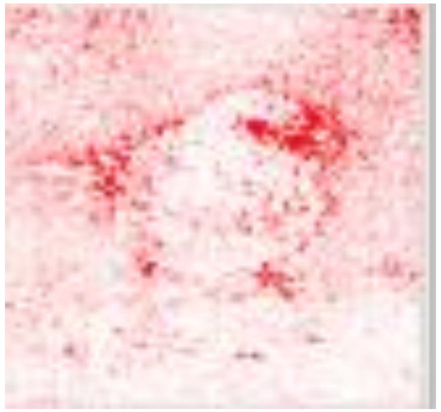
- Gradients can get saturated
- Solution: Take (element-wise) product of the gradient with the input itself
- Can produce visually simpler/sharper images
- Can be used in conjunction with any method

Shrikumar et al., Not just a black box: Learning important features through propagating activation differences, ICML’17

Gradient-Input



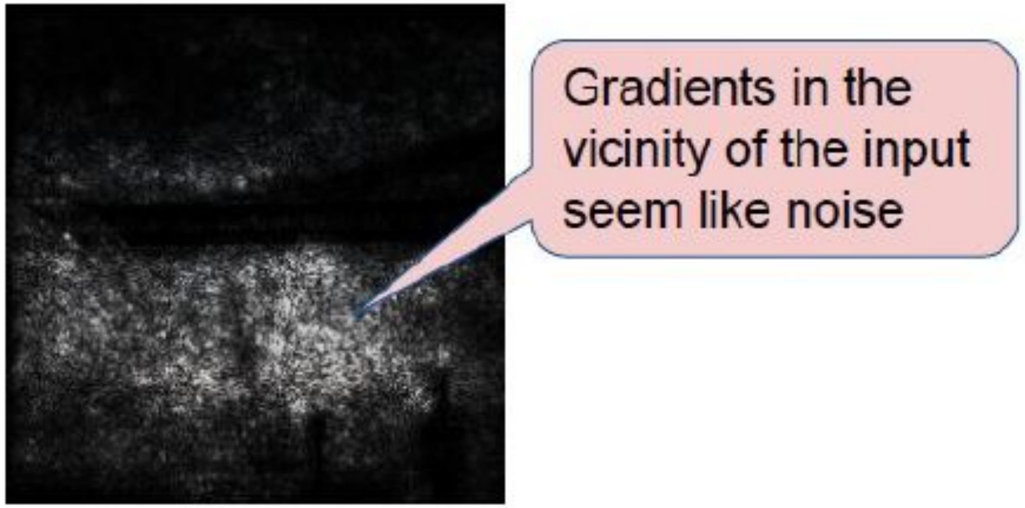
$$\nabla_x F(x) \odot x$$



Element-wise product of input-gradient and input

Integrated Gradients

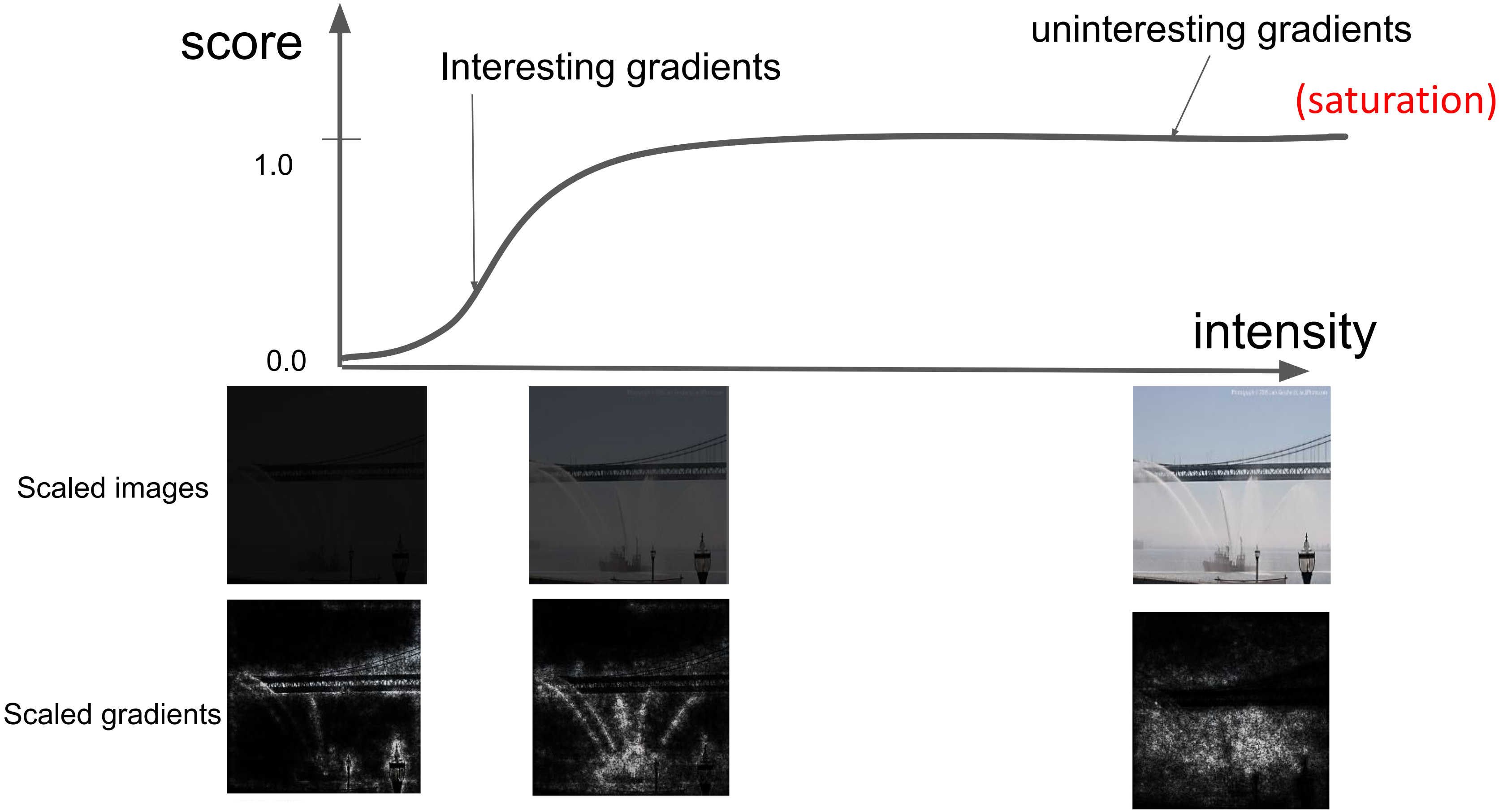
- Input gradients indicate influence
- However, gradient for actual input may not give the full picture



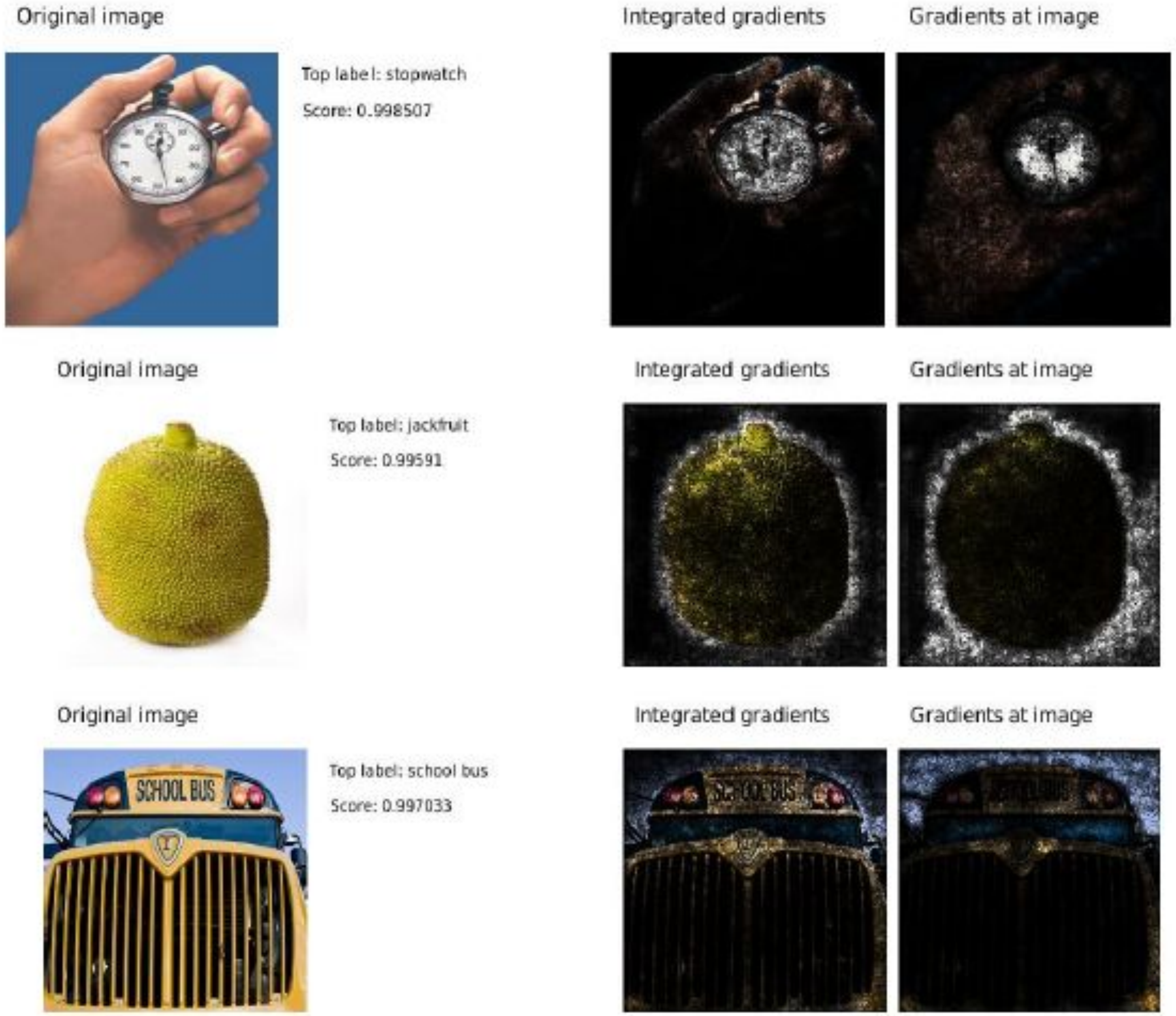
Sundararajan et al., Axiomatic Attribution for Deep Networks, ICML’17

Integrated Gradients

<u>Solution:</u> integrate gradients over a range of possible inputs between *baseline* input and given input



Examples of improved explanations



Integrated Gradients: Key ideas

- ***Baseline input:*** A baseline is an uninformative input used for comparison
    - E.g. Object recognition: black image, noise image

- List of **desirable criteria (axioms)** for an attribution method
    - ***Implementation Invariance Axiom:*** Two networks that compute identical functions for all inputs get identical attributions even if their architecture/parameters differ
    - ***Sensitivity Axiom:***
        - If baseline and input have different scores, but differ in a single variable, then that variable gets some attribution.
        - If a variable has no influence on a function, then it gets no attribution.
    - ***Linearity Preservation Axiom:*** Attributions(a*f1 + b*f2) = a*Attributions(f1) + b*Attributions(f2)
    - ***Completeness Axiom:*** sum(attributions) = f(input) - f(baseline)
    - ***Symmetry Preservation Axiom:*** Symmetric variables with identical values get equal attributions

- Integrated Gradients is the only method that satisfies these desirable criteria

Sanity checks for saliency maps

- How do we evaluate the validity of different techniques for generating saliency maps?
  - Do not have ground truth!
  - Human judgment can be biased!

- Use “sanity checks” for saliency maps
  - Model parameter randomization test: What happens to saliency maps when model parameters are randomized?
  - Data randomization test: What happens to saliency maps when model is trained on randomly labeled data?

Adebayo et al., Sanity Checks for Saliency Maps, NeurIPS’18

Sanity checks for saliency maps

- Results:
  - Methods such as Guided Backpropagation and Guided GradCAM fail sanity tests
  - Methods such as vanilla Gradient, SmoothGrad pass the the sanity tests

- Reasons for sanity test failure:
  - Random DNNs may still demonstrate meaningful behavior because of architectural bias
  - Methods for generating saliency maps have over-optimized for desirable visual effects

Questioning assumptions

- Does the gradient wrt the input really inform us about the most important features?
- How robust are interpretations to small perturbations?

Ghorbani et al., Interpretation of Neural Networks is Fragile, AAAI’19

Srinivas et al., Rethinking the Role of Gradient-Based Attribution Methods for Model Interpretability, ICLR’21

Shah et al., Do Input Gradients Highlight Discriminative Features?, NeurIPS’21

Adebayo et al., Post hoc Explanations may be Ineffective for Detecting Unknown Spurious Correlation, ICLR’22

Bilodeau et al., Impossibility theorems for feature attribution, PNAS’24